Rational Number Project

Initial Fraction Ideas Lesson 2: Overview	Materials
Students explore relationships among circle pieces, modeling and orally naming fraction amounts for: 1-half and 1-third.	• Fraction Circles for students and teacher • Student Page A

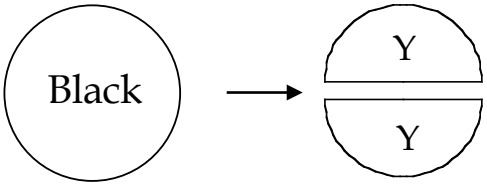
Teaching Actions

Warm Up

Find two different ways to cover 1 yellow piece.
Find two different ways to cover 1 brown piece.

Large Group Introduction

1. Ask students to take out a black circle. Model how to divide the black circle into 2 equal parts by showing that 2 yellow parts cover the whole circle.



[2 yellows cover 1 black circle]

2. Ask: How many yellows equal the black circle? Are the 2 parts covering the black circle equal?

3. Explain: the black circle is the unit or the whole you partitioned into two equal parts. When 2 equal parts equal one *unit* or *whole*, each part (pick up 1 yellow) is called one-half. This yellow piece is one-half of the black circle.

4. Show one-half by placing 1 yellow on the black circle. *"1 yellow covers half of the black circle."*

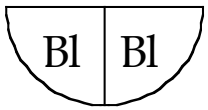
Comments

Most lessons will have a "Warm Up" problem that reviews ideas from previous lessons. These warm ups are not meant to take more than 5 minutes of class time.

Flexibility of unit is stressed right from the beginning by having students find multiple representations for 1/2, 1/3, and 1/4.

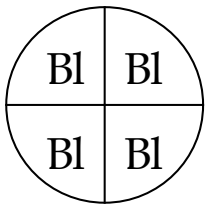
The critical variable with fractions is that a unit or a whole is divided into equal parts. A single part can be given a fraction name that depends on what it is being compared to.

Example:



2 blues equal 1 yellow so 1 blue is one-half of the yellow. [Here 1 yellow is the unit]

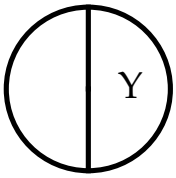
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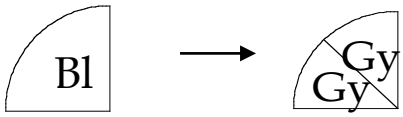
4 blues equal 1 black. 1 blue is one-fourth of the black circle. [Here black is the unit]

Students confuse the terms unit and whole. Sometimes students will use the word *whole* for the object partitioned and use the term *units* to

Teaching Actions



5. Continue by looking for other examples for one-half. Show one blue piece and ask: This one blue is my unit or whole that is to be divided. How can we divide this piece into 2 equal parts? What color pieces will do this?

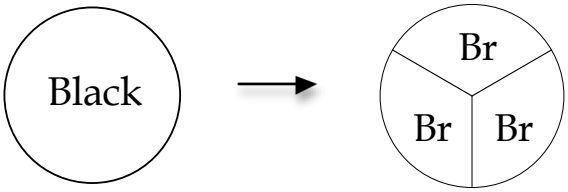


6. Use these questions: Are the 2 parts equal? 1 gray is 1 of 2 equal parts; what fraction of the blue piece is 1 gray? [1-half]



7. Show 1 yellow and ask students to consider the yellow as the unit or the whole to be divided. Which piece divides the yellow into two equal parts? What name can we give to each piece? What fraction of the yellow is each piece?

8. Model, using the black circle as the unit or the whole, representations for thirds.



9. 3 browns cover 1 black; 1 brown is 1 of 3 equal parts; 1 brown is one-third of the black. Show as:

Comments

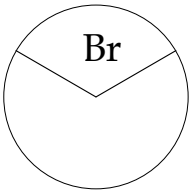
name the equal parts. But the unit is another name for the whole. The object being partitioned is the unit or whole; to name a fractional amount students compare how many equal parts make up the unit/whole. For example, because 4 blues partition the black circle into 4 equal parts, 1 blue is ¼ of the black circle. But 1 blue is half of the yellow (2 blues cover 1 yellow) if you consider one yellow piece as the unit or whole that is partitioned.

Students are naming fractions in the verbal mode only. In the lesson 3 students will record as: 1-fourth.

You may want to show how other units can be partitioned into halves:

1 pink → 2 reds
1 brown → 2 pinks
1 orange → 2 purples

Teaching Actions



10. Find other examples for 1-third using 1 yellow and 1 brown and then 1 blue as the unit.

Small Group/Partner Work

11. Student page A presents problems similar to ones presented in large group. Assign to students in pairs, as they are to name the fractions verbally.

Wrap Up

12. To assess the "big idea" in this lesson present the following scenario display one of each of these colors: yellow, blue, pink, and red on the and ask: "You have called all of these 1-half, yet they are different sizes. How is that possible?"

Comments

Translations

- Manipulative to verbal
- Pictures to verbal

Find two different ways to cover 1 yellow piece.

Find two different ways to cover 1 brown piece.

Name_____ Grade 3 Lesson 2 / Student Page A

Work with a partner to complete this activity. Use your fraction circles. Say the fraction name to each other when asked.

1. Take out one yellow piece. This is the unit or the whole you will divide into equal parts.

   How many blues cover one yellow piece? _____

   1 blue is _____ of one yellow.

   (Say the fraction name)

2. Take out one brown piece. This is the unit or the whole you will divide into equal parts.

   How many pinks cover one brown piece? _____

   1 pink is _____ of one brown.

   (Say the fraction name)

3. One yellow piece is the unit or whole you will divide into equal parts.

   How many pinks cover one yellow piece? _____

   1 pink is _____ of one yellow.

   (Say the fraction name)

4. What color piece is 1-half of one blue? _____

5. What color piece is 1-third of one yellow? _____

6. What color piece is 1-half of one black circle?_____

7. What color piece is 1-third of one black circle?_____

8. What color piece is 1-third of one orange piece?_____

9. What color piece is 1-half of one pink piece?_____